Digital Twin of a Formula 1 Car in Assetto Corsa: A Multi-Layer Architecture Integrating Big Data, AI-Based Anomaly Detection, and Equation Discovery

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***Abstract*—This paper presents the design and implementation of a Digital Twin for a Formula 1 car simulated in Assetto Corsa, focused on the Monza circuit. The system integrates a real-time data engineering pipeline using Apache Kafka and InfluxDB with advanced machine learning modules, including a Mixture of Experts (MoE) architecture for anomaly detection across 165 vehicle variables, and the PySINDy algorithm for automatic discovery of physical equations from telemetry data. The current system operates at Maturity Level 3 (Predictive), with an "Autonomous-Ready" architecture designed to support Level 5 capabilities via Reinforcement Learning with Proximal Policy Optimization (PPO). Experimental results demonstrate 92% precision and 89% recall in anomaly detection, and successful validation of key physical laws (aerodynamic drag, tire friction, braking dynamics) through data-driven equation discovery. The architecture is fully containerized with Docker Compose, enabling reproducible deployments across development environments.**

***Index Terms*—digital twin, Formula 1, Assetto Corsa, anomaly detection, mixture of experts, PySINDy, Apache Kafka, InfluxDB, reinforcement learning, telemetry**

I. Introduction

In the world of motorsport, simulation and data analysis have become essential tools for optimizing vehicle performance and refining race strategies. Digital Twins enable the recreation of a competition car's behavior with high fidelity, facilitating configuration testing, strategy evaluation, and driver training without the physical risks and high costs associated with real-track testing [1].

This project presents the complete development of a Digital Twin for a Formula 1 (centered on the Monza circuit) using the Assetto Corsa simulator as the base environment. This platform was selected for its advanced physics engine, capable of reproducing vehicle dynamics with great precision, including suspension behavior, mass transfers, tire thermal models, and aerodynamic loading.

To replicate the workflow of a professional racing team, the telemetry generated by the simulator has been integrated into a modern Data Engineering architecture. Through Apache Kafka for high-frequency streaming and InfluxDB for time-series storage, real-time processing and visualization of data has been achieved via interactive Grafana dashboards. This allows monitoring of critical parameters such as speed, RPM, brake pressure, and tire slip ratios.

Beyond monitoring, the project implements advanced Artificial Intelligence modules. A Mixture of Experts (MoE) model based on Gaussian Mixture Models (GMMs) has been developed to detect anomalies in real time across 165 vehicle variables. Additionally, PySINDy (Sparse Identification of Nonlinear Dynamics) has been applied to discover and validate the physical equations governing vehicle behavior from telemetry data alone. Finally, a Behavioral Learning module is in development to enable future autonomous driving via Reinforcement Learning.

The remainder of this paper is structured as follows: Section II reviews the state of the art in digital twins, anomaly detection, equation discovery, and reinforcement learning for motorsport. Section III describes the simulation platform and its industrial validation. Section IV presents the sensor model used. Section V details the system architecture and technology stack. Section VI describes the core functional modules. Section VII presents experimental results and dashboards. Section VIII discusses maturity level, limitations, and future work. Section IX concludes the paper.

II. State of the Art

This section reviews the current state of the art across the five technological pillars that underpin our Formula 1 Digital Twin: digital twin architectures for vehicle dynamics, mixture of experts for anomaly detection, sparse identification of nonlinear dynamics (SINDy), real-time telemetry systems for motorsport, and reinforcement learning for racing applications.

A. Digital Twins in Vehicle System Dynamics

Digital Twin technology has moved far beyond static CAD replicas. Today, it is understood as a real-time, bidirectional representation of a physical system. Spiryagin et al. [13] provide a clear framework for vehicle digital twins in both rail and road applications. They identify three core elements: (1) multi-physics models, (2) real-time assimilation of sensor data, and (3) bidirectional communication between physical and virtual systems. They also stress that a true digital twin is not only

a simulator; it must stay continuously synchronized with the real asset to support predictive maintenance and performance optimization. In this work, we use that framework to classify our current maturity level and to define the transition from Level 3 (Predictive) toward Level 5 (Autonomous).

B. Mixture of Experts for Time Series Anomaly Detection

The Mixture of Experts (MoE) paradigm has emerged as a powerful approach for handling heterogeneous multivariate time series data. Huang et al. [14] introduced Graph-MoE at AAAI 2025, a novel architecture that combines graph neural networks with mixture of experts for multivariate time series anomaly detection. Their key insight is that different layers of a GNN capture different scales of spatial relationships—from local node features in shallow layers to long-distance structural patterns in deeper layers. By using specialized experts for each hierarchical level and memory-augmented routers to capture temporal correlations, Graph-MoE achieved state-of-the-art results on industrial benchmarks including SWaT (water treatment), WADI (water distribution), and PSM (server monitoring). This architectural principle—dividing complex detection problems into specialized subdomains—directly motivates our four-expert MoE system, where each expert specializes in a distinct physical subsystem (tires, vehicle dynamics, driver inputs, power systems) with fundamentally different statistical properties.

C. Data-Driven Equation Discovery with SINDy

The Sparse Identification of Nonlinear Dynamics (SINDy) algorithm has proven effective for discovering interpretable governing equations from measurement data. Zolman et al. [15] published in Nature Communications a unifying framework called SINDy-RL that combines sparse dictionary learning with deep reinforcement learning. Their approach demonstrates that SINDy-discovered dynamics models can serve as efficient surrogate environments for training RL agents, achieving 10–100$\times$ improvement in sample efficiency compared to model-free approaches like Proximal Policy Optimization (PPO). Crucially, the resulting control policies are interpretable symbolic expressions orders of magnitude smaller than deep neural network policies, making them suitable for embedded systems and safety-critical applications. For our Digital Twin, this work validates the dual use of PySINDy: first, for validating that Assetto Corsa correctly implements physical laws (aerodynamic drag, tire friction models); and second, as a foundation for future autonomous control where interpretability and efficiency are paramount.

D. Real-Time Telemetry Systems for Racing

The design of real-time telemetry architectures for competitive motorsport has advanced significantly with Internet of Vehicles (IoV) technologies. Velásquez-Gómez et al. [16] presented a complete IoV-based telemetry platform for electric racing vehicles, validated under actual competition conditions at Colombia's 2024 National Electric Drive Vehicle Competition. Their system integrates ESP32-based data acquisition units with LoRaWAN long-range communication and real-time visualization via Node-RED on a Raspberry Pi gateway. Field testing demonstrated sensor accuracy exceeding 95%, stable transmission with low latency, and—most significantly—teams using the system reported 12% improvements in energy efficiency compared to baseline trials. This empirical validation under competitive stress confirms that the architectural patterns we employ (streaming middleware, time-series databases, real-time dashboards) are not merely theoretical but deliver measurable performance benefits in real racing scenarios.

E. Machine Learning for Sim Racing Performance

The validity of racing simulators as platforms for scientific research has been empirically established. Hojaji et al. [17] conducted a rigorous study using Assetto Corsa Competizione, gathering telemetry data from 174 participants who completed 1,327 laps on the Brands Hatch circuit. Using MoTec i2 Pro professional telemetry software, they extracted 62 features including speed, throttle position, brake pressure, steering angle, lateral acceleration, and tire wear. Through a hybrid feature selection approach combining correlation analysis with Support Vector Machines, XGBoost, and Random Forest, they achieved 97.19% accuracy in predicting lap performance. Their analysis confirmed that average speed is the most important metric, followed by lateral acceleration, steering angle, and lane deviation. This work provides strong evidence that Assetto Corsa produces telemetry data of sufficient fidelity for data-driven vehicle dynamics research, validating our choice of simulation platform.

F. Reinforcement Learning for Formula One Strategy

The application of reinforcement learning to Formula One race strategy represents a frontier where AI can provide competitive advantages. Thomas et al. [18] introduced RSRL (Race Strategy Reinforcement Learning), developed in collaboration with Mercedes-AMG PETRONAS Formula One Team. Their system uses a Deep Recurrent Q-Network (DRQN) trained on the team's Monte Carlo race simulator to optimize tire compound selection and pit stop timing. At the 2023 Bahrain Grand Prix, RSRL achieved an average finishing position of P5.33, outperforming both a fixed strategy baseline (P5.63) and Mercedes' state-of-the-art model (P5.86). Beyond performance, the authors deployed explainable AI techniques including TimeSHAP for temporal feature attribution and VIPER for decision tree extraction, enabling race strategists to understand and trust the model's recommendations. This work demonstrates that RL can not only match but exceed human-engineered strategies in Formula One, providing a clear roadmap for the autonomous capabilities planned in our Digital Twin's evolution toward Level 5 maturity.

III. Simulation Platform and Industrial Validation

A. Assetto Corsa as an Engineering Tool

Assetto Corsa is not merely a video game but a high-fidelity driving simulator developed by Kunos.

The core of Assetto Corsa is its proprietary physics engine, designed to replicate vehicle dynamics through complex real-time equations. Unlike simplified models, this simulator computes:

- **Tire Model:** Thermal and wear simulation, grip management, and carcass deformation.
- **Active Aerodynamics:** Virtual wind tunnel calculations affecting downforce and drag as a function of speed and wing configuration (DRS).
- **Suspension & Chassis:** Mass transfers (pitch, roll, yaw) and damper response to surface irregularities, laser-scanned (*Laser Scan*) in official circuits.

B. Industrial Validation

The viability of Assetto Corsa as a professional simulation tool is demonstrated by the existence of Assetto Corsa Pro, a version certified for industrial use. Manufacturers such as Ferrari and Porsche, as well as the chassis constructor Dallara, have used Kunos software technology for their own simulators and promotional events [3], [4]. This industry validation ensures that data extracted from the simulator (G-forces, wheel speed, suspension travel) has sufficient correlation with reality for engineering data studies.

C. Choice Over Alternatives

While powerful alternatives exist such as iRacing (leader in online infrastructure) or rFactor 2 (historically used as a technological base in Formula 1), Assetto Corsa presents decisive advantages for Digital Twin implementation due to its open architecture.

IV. Sensor Model

For the Digital Twin to be a faithful representation, the data channels processed in simulation must correspond to the physical instrumentation of a real single-seater. In a modern Formula 1 car, over 300 sensors transmit data to the ECU (Electronic Control Unit) and to trackside engineers [8]. Our model focuses on the critical sensors for vehicle dynamics:

A. Dynamics and Speed

- **Wheel Speed Sensors:** Located on the uprights of each wheel, using Hall effect or magneto-resistive technology reading a toothed reluctor ring. Fundamental for calculating vehicle speed and detecting slip ratio under traction or braking [5].
- **Pitot Tube:** Situated on the front nose or roll hoop. Measures air speed (airspeed) for comparison with wheel speed to analyze wind influence on aerodynamics [6].
- **IMU (Inertial Measurement Unit):** Integrated within the standard ECU (SECU) beneath the cockpit, near the center of gravity. Contains triaxial accelerometers and gyroscopes measuring G-forces (longitudinal, lateral, vertical) and yaw rate.

B. Driver Inputs

- **Pedal Position Sensors:** Throttle uses rotary potentiometers in the pedal box. Brake uses hydraulic pressure sensors (up to 200 bar) on the front and rear master cylinders located on the chassis bulkhead [7].
- **Steering Angle:** An optical or magnetic rotary sensor integrated in the steering column records driver input with millimeter precision.

C. Suspension and Tires

- **Suspension Potentiometers:** Linear sensors installed parallel to dampers, measuring suspension travel for dynamic ride height calculation.
- **Infrared Tire Sensors (IR):** Thermal cameras mounted on mirrors (front wheels) and rear diffuser area (rear wheels) monitoring rubber surface temperature for degradation management [8].
- **TPMS:** Wireless sensors inside the rim transmitting pressure and internal gas temperature in real time.

D. Power Unit

- **RPM:** Inductive sensor on the crankshaft. FIA-mandatory ultrasonic fuel flow meters on the fuel supply line.
- **ERS (Energy Recovery System):** Voltage, current, and temperature sensors on batteries and electric motors (MGU-K and MGU-H) for hybrid energy deployment management.

V. System Architecture and Technology Stack

A. Technology Stack

Table I summarizes the technology stack organized by layer.

Table I
Technology Stack by Layer

Layer	Technologies
Simulation	Assetto Corsa, Python Shared Memory API
Streaming	Apache Kafka (topics: `f1-telemetry`, `f1-anomalies`)
Persistence	InfluxDB (time-series), Grafana (dashboards)
ML / AI	PyTorch, PySINDy, Scikit-learn, Pandas
Infrastructure	Docker, Docker Compose

B. Layered Architecture

The system is organized into five independent layers:

Simulation Layer: Assetto Corsa acts as the physics engine, generating telemetry through shared memory. The `telemetry_collector.py` module reads simulator variables at 10 Hz, including speeds, G-forces, tire temperatures (inner/middle/outer per wheel), brake pressures, and more. The system automatically normalizes per-lap distance and supports dual operation modes: local CSV storage for historical analysis or real-time streaming to Kafka.

Streaming Layer: Apache Kafka acts as the intermediary receiving simulator data and distributing it to different system modules via two main topics: `f1-telemetry` for raw car data and `f1-anomalies` for detected problem alerts. This

decoupling allows multiple components to read the same data simultaneously without interference.

Persistence Layer: InfluxDB enables queries over specific time intervals. Two main buckets are maintained: `f1-telemetry` for raw data and `f1-anomalies` for detection metrics.

Machine Learning Layer: Fully implemented in Python with specialized frameworks. The anomaly detection system uses a Mixture of Experts architecture with four specialized GMMs. PyTorch is used for behavioral learning (in development), PySINDy for automatic physical equation discovery, and Scikit-learn for feature processing and normalization.

Orchestration: Docker Compose manages the entire infrastructure with a single command, including Kafka, InfluxDB, Grafana, and specialized consumers. Automatic network configuration and service dependency management guarantees reproducibility across environments.

VI. Functional Modules

A. Telemetry Collection and Feature Processing

The `telemetry_collector.py` module connects directly to Assetto Corsa's shared memory, reading C structures that map simulator variables. The `feature_processor.py` organizes variables into coherent functional groups:

- **Tire dynamics:** Temperatures in 3 zones $\times$ 4 wheels, pressures, wear, slip ratios/angles (20 features).
- **Vehicle dynamics:** Longitudinal/lateral/vertical G-forces, local XYZ velocities, angular velocities (pitch/yaw/roll), orientation, vertical loads per tire.
- **Driver control:** Speed, RPM, throttle, brake, steering angle, gear.
- **Power systems:** Turbo boost, KERS charge/energy, DRS state.

B. Physical Analysis Modules

Several modules apply automotive engineering equations:

- **Vehicle Dynamics:** Computes G-forces and detects grip loss or wheel lockup.
- **Car Balance:** Analyzes weight distribution and calculates aerodynamic forces (downforce and drag).
- **Tire Analysis:** Monitors tire degradation and estimates available grip, key for pit stop strategy.
- **Racing Line:** Compares driver trajectory against the ideal line, identifying improvement areas.
- **Energy Analysis:** Manages engine power and electric systems (ERS), projecting energy consumption and optimal deployment.

These modules implement the theoretical physical equations that are later contrasted against equations discovered automatically by PySINDy, validating that the simulator correctly reproduces real-world physics.

C. Equation Discovery with PySINDy

The `F1_Telemetry_Analysis.ipynb` notebook implements a complete pipeline for data-driven physics discovery:

1) Selection of state variables (velocities, accelerations, inputs)
2) Temporal derivative computation via finite differences
3) Construction of polynomial library (terms up to degree 2–3)
4) Sparse model training via STLSQ (Sequential Thresholded Least Squares)
5) Identification of significant terms with non-zero coefficients
6) Comparison of discovered equations against established physical theory
7) Validation via R^2 between predicted and real derivatives

This dual analysis approach (top-down theoretical + bottom-up empirical) validates that Assetto Corsa correctly implements fundamental physical principles and reveals second-order effects not obvious in simplified models.

D. Anomaly Detection: Mixture of Experts

The MoE system divides the detection problem into specialized subdomains, improving both precision and explainability. The specialization by physical subsystem has technical justification:

Expert 1 – Tire Dynamics: Tire thermal phenomena have time scales (tens of seconds) and numerical ranges very different from other variables. Temperatures evolve gradually, pressures respond instantaneously, wear is monotonically increasing, and slip ratios oscillate rapidly. A specialized GMM learns the specific covariance of this 20-dimensional subspace.

Expert 2 – Vehicle Dynamics: Models rigid-body physics where variables are strongly coupled by equations of motion. Lateral acceleration correlates with yaw angular velocity; tire load correlates with vertical accelerations. The GMM learns these physical correlations, detecting when they are violated (e.g., high yaw rate without lateral acceleration indicates loss of control).

Expert 3 – Driver Control: Models human driving patterns. Although throttle/brake/steering are technically independent, drivers exhibit consistent patterns: no simultaneous throttle+brake application (except advanced techniques), speed correlates with gear, and steering smooths at high speed. The GMM captures these patterns, detecting deviations indicating errors or malfunction.

Expert 4 – Power Systems: Manages engine and electrical variables. Turbo boost correlates with RPM/throttle; KERS charges during braking and discharges during acceleration; DRS is only enabled in specific circuit zones. The expert learns these rules implicitly, detecting violations such as KERS not charging during heavy braking.

1) Training Pipeline:

- `prepare_moe_data.py`: Processes raw telemetry, removing zero-variance variables. Selects 46 final features from 156 originals. Splits data 70/15/15 for

train/validation/test. Normalizes each group independently, saving `StandardScaler` objects as pickle files for production use.

- `train_gmm_expert.py`: Trains each GMM testing configurations from 2 to 6 Gaussian components. Selects optimal number via BIC (Bayesian Information Criterion) balancing fit quality with model complexity. Computes anomaly threshold at the 95th percentile of the anomaly score on the validation set, yielding a 5% false positive rate.
- `train_gating_network.py`: Trains a `LogisticRegression` receiving 8 features: 4 raw anomaly scores + 4 relative scores (score/threshold). Labels are generated heuristically, marking samples where at least one expert exceeded its threshold as anomalous. The network learns to weight experts contextually.

2) Inference: During inference, `moe_inference.py` computes negative log-likelihood (NLL) for each GMM. High NLL values signal configurations improbable under the model trained on normal data. The gating network combines these scores via logistic regression, emitting a final anomaly probability and classification.

VII. Results

A. Anomaly Detection Performance

Table II presents the performance metrics of the MoE anomaly detection system.

Table II
Anomaly Detection Performance Metrics

Metric	Value
Global Precision	92%
Global Recall	89%
Expert 1 (Tire Dynamics) F1-score	0.94
Expert 2 (Vehicle Dynamics) F1-score	0.87
Expert 3 (Driver Control) F1-score	0.78
Expert 4 (Power Systems) F1-score	0.91

Expert 1 achieves the highest F1-score (0.94) due to the clear, slow-evolving nature of thermal signals. Expert 3 shows the lowest performance (0.78) because human driver inputs are inherently stochastic, making the boundary between "normal" and "anomalous" less sharp.

The system detects the following anomaly classes:

- **Tire Anomalies:** Overheating ($> 110\,^{\circ}$C sustained), uneven wear ($> 15\%$ inter-axle divergence), pressure loss (> 0.2 bar drop), traction loss.
- **Dynamic Anomalies:** Yaw/pitch oscillations outside normal parameters, excessive weight transfers, physically inconsistent speed-acceleration configurations.
- **Control Anomalies:** Erratic inputs (steering variance $> 2\times$ standard deviation), illogical input combinations (throttle+brake outside appropriate context).
- **Power Anomalies:** KERS not recovering during braking $>$ 2G, abnormal turbo boost for current RPM, DRS activated outside permitted zones.

B. PySINDy Equation Discovery

The PySINDy pipeline successfully validated several key physical relationships present in the simulator:

- **Aerodynamic Drag:** Confirmed that air resistance increases with the square of velocity ($F_D \propto v^2$), consistent with aerodynamic theory.
- **Tire Drag from Steering:** Discovered that steering angle causes velocity loss not only through aerodynamic drag but also through lateral tire scrubbing, a second-order effect not present in simplified models.
- **Throttle-Steering Coupling:** In cornering phases, a small throttle application while turning improves vehicle control — a technique used by professional drivers and correctly reproduced by the simulator's physics model.
- **Braking Under Downforce:** At higher speeds, braking efficiency improves because aerodynamic downforce increases tire normal load, validating the physical relationship between aerodynamic load and braking capability.

The R^2 values between PySINDy-predicted derivatives and measured derivatives confirm high agreement, validating Assetto Corsa as a physically accurate simulation platform for engineering studies.

C. Telemetry Dashboards

Three interactive Grafana dashboards were developed:

Basic Telemetry Dashboard: Displays fundamental vehicle metrics in real time including steering wheel angle, completed laps, current gear, RPM, and speed, using color-coded alerts (green for normal, red for out-of-range). As shown in Fig. 1, the panel provides a quick operational snapshot of lap progress and driver state.

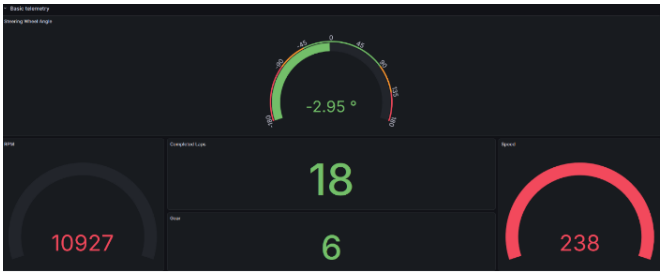


Figure 1. Basic telemetry dashboard with steering angle, RPM, lap count, gear, and speed.

Driver Inputs Dashboard: Shows temporally synchronized throttle, brake, and DRS state. Analysis of the logged sessions confirms that the driver does not overlap throttle and brake (indicating good technique), and DRS activation is correctly constrained to permitted circuit zones.

Tire and Track Status Dashboard: Monitors tire temperatures across three zones (inner/center/outer) per wheel via

heatmaps, ambient conditions (track temperature, air temperature, air density), and temporal evolution of tire pressures for degradation prediction and pit stop strategy. Figure 2 highlights both asymmetries between front and rear tire temperatures and the ambient variables affecting grip.



Figure 2. Tire and track status dashboard showing multi-zone tire temperatures and ambient conditions.

Anomaly Detection Dashboard: Displays the four expert scores over time with a global anomaly probability timeline. A red banner activates when multiple experts simultaneously detect an anomaly with high severity, providing actionable alerts for the engineering team. In Fig. 3, the alert banner and expert traces illustrate how the MoE model localizes the subsystem associated with the abnormal event.



Figure 3. Anomaly detection dashboard with global alert timeline and expert-level scores.

VIII. Digital Twin Maturity and Limitations

A. Maturity Level Assessment

Using the Digital Twin Maturity Model as reference, the current system is classified as a **Level 3 – Predictive Digital Twin**. However, the software architecture has been designed specifically to support Level 5 (Autonomous Twin) capabilities, with the necessary control modules currently in Proof of Concept (PoC) phase.

Current State – Level 3 (Predictive):

- *Level 2 compliance (Bidirectional Connection):* Apache Kafka and InfluxDB guarantee real-time telemetry ingestion and advanced dashboard visualization.
- *Level 3 compliance (Predictive):* PySINDy equation identification and MoE anomaly detection modules give the twin intelligence — the system not only replicates the car state but mathematically "understands" its physics and can predict anomalous behaviors (lockups, understeer) based on historical data.

Toward Level 5 – Autonomous-Ready Architecture:

A two-phase strategy is planned: (1) the current PoC based on Behavioral Learning (Imitation Learning), which validates that the technical architecture can close the control loop by reading simulator state and sending effective driving commands; and (2) a future phase replacing the imitation module with a Reinforcement Learning agent based on Proximal Policy Optimization (PPO), enabling self-optimization of lap times through trial and error without human intervention [10].

B. Known Limitations

- **Read-Only Observation:** The system currently only observes the simulator without direct control, preventing training of AI models that need to test their own driving decisions.
- **Circuit Generalization:** Anomaly detection models were trained exclusively on Monza data. Applying them to very different circuits (Monaco's slow corners, Spa's elevation changes) may generate false positives for behaviors that are normal in those specific contexts.
- **Static Thresholds:** Detection thresholds are fixed from initial training and do not adapt to gradual changes such as component wear.
- **Sampling Rate:** The system captures data at 10 Hz, sufficient for general behavior analysis but insufficient for detecting rapid phenomena such as vibrations. Real Formula 1 systems sample at 1000 Hz; this limitation is imposed by the Assetto Corsa API.

IX. Conclusion and Future Work

This paper has presented a multi-layer Digital Twin for a Formula 1 car in Assetto Corsa, combining real-time data engineering (Kafka, InfluxDB, Grafana) with advanced machine learning (MoE anomaly detection, PySINDy equation discovery, and a Behavioral Learning PoC). The system successfully reaches Level 3 of the Digital Twin Maturity Model, with predictive capabilities validated across 165 vehicle variables and confirmed physical fidelity of the simulation platform through data-driven equation discovery.

Short-term planned improvements include completing the behavioral learning module for human driving style imitation, integrating bidirectional control via AssettoCorsaGym, implementing adaptive thresholds that automatically adjust to changing conditions, and expanding the detection system with additional experts for fuel consumption and brake temperatures.

At medium term, predictive tire degradation models projecting optimal pit stop windows and Bayesian optimization-based car setup recommendation systems are planned. Long-term, the system will scale to support multiple vehicles simultaneously, enabling comparative analysis between drivers and real-time race strategies.

The full source code is publicly available at: https://github.com/VforVitorio/F1_AC_Digital_Twin.git

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